

SECTION 5

PERFORMANCE DATA

5.1 GENERAL

This section contains the performance information necessary for preflight and inflight planning. The information is primarily intended for mission planning and is most useful when planning operations in unfamiliar areas or at extreme conditions. Although maximum performance is not always required, regular use of the information in this section is recommended for the following reasons:

- Knowledge of performance margins will allow better decisions when unexpected conditions or alternate missions are encountered.
- Situations requiring maximum performance will be more readily recognized.
- Familiarity with the data will allow performance to be calculated more easily and quickly.
- Experience will be gained in accurately estimating the effects of conditions for which data is not presented.

5.1.1 Definitions

Some terms which are used particularly in the performance section are defined as follows:

5.1.1.1 Airspeeds

IAS The indicated airspeed is the reading obtained on a pitot-static airspeed indicator having no instrument calibration error. If the calibration error on the particular instrument is not known, then the actual reading may be taken equal to IAS because the tolerances permitted on the instrument are small.

CAS The calibrated airspeed is the IAS-value corrected for pitot-static position errors only.

TAS The true airspeed of the helicopter relative to undisturbed air. It can be obtained by multiplying the CAS value by the true airspeed factor, if compressibility is neglected.

V_{NE} The never exceed speed, IAS.

V_Y The speed for best rate of climb, IAS.

5.1.1.2 Height

The true vertical clearance between the lowest part of the helicopter and the relevant datum.

5.1.1.3 Outside Air Temperature (OAT)

The ambient temperature of the free airstream, expressed in °C.

5.1.1.4 Pressure Altitude

The expression of atmospheric pressure in terms of altitude, according to the interrelation of these factors in the international standard atmosphere. Pressure altitude may be obtained by setting the subscale of an accurate barometric altimeter to 1013.2 hPa (29.92 inches or 760 mm of mercury).

NOTE All altitudes given in this manual are pressure altitudes, unless otherwise indicated.

5.1.2 Standard Performance Conditions

The data presented cover the maximum range of performance that can be expected and are based on the following conditions:

- Engine power does not exceed limits (see Section 2).
- Installation and accessory losses are included.
- Rotor RPM - 100 %.

In each area of performance, the effects of altitude, temperature, gross mass, and other parameters relating to that phase of flight are presented.

5.1.3 Conditions Affecting Performance

Conditions affecting performance are listed in the heading of the respective charts. The data presented are accurate only for these specific conditions.

Other general conditions applicable to the performance data are:

- Airframe and engine control rigging (assumed within allowable tolerances).
- Pilot technique (assumed normal, control movements smooth and continuous).
- Clean performance influencing components (rotor blades, engine air intakes and compressor sections).

5.1.4 Reading the Charts

None of the curves presented should be extrapolated, but interpolation between given data is permissible.